Day 6 – DroidCam Integration & Stabilization

Date: 16/02/2026

Duration: Approximately 1 hour 30 minutes

Objective of Day 6

- Make DroidCam work reliably with the Python/OpenCV setup.
- Eliminate green screen and device busy errors.
- Prepare a stable camera setup for dataset collection.

1. What Was Done

- Enabled Developer Mode on the Android phone.
- Connected the phone to the laptop using a USB charging cable.
- Configured DroidCam to operate through USB instead of WiFi.
- Ran the Python OpenCV script successfully.
- Confirmed that the camera feed opened through DroidCam.
- Resolved previous backend and device locking issues.

2. Challenges Faced

- Backend conflicts when accessing DroidCam.
- Device locking issues where only one application could use the camera.
- WiFi streaming instability.
- Limited USB cable quality for high-speed video transmission.

3. Lessons Learned

- USB connection is more stable than WiFi for video capture.
- Only one program can access a webcam device at a time.
- Most errors were hardware/driver related, not MediaPipe or ML logic.
- Stabilizing hardware before data collection prevents wasted effort.

4. Plan for Day 7

- Begin collecting landmark data for Ethiopian Sign numbers.
- Start with a small batch (0–5) to validate consistency.
- Maintain consistent lighting and hand positioning.

- Organize dataset folders properly for ML training.